

Positional Encoding

Deep Learning

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Contents of This Video

In this video, we will cover:

- Why attention needs position information
- Sinusoidal positional encoding
- How encoding captures position
- Learned vs. fixed positional encodings
- Adding position to input embeddings

The Problem: Attention is Position-Blind

Self-attention computes:

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^\top}{\sqrt{d_k}} \right) V$$

If we shuffle inputs, attention shuffles the same way

“The cat sat” and “sat cat The” → same relationships!

But word order matters in language:

- “Dog bites man” $\neq$ “Man bites dog”

Sinusoidal Positional Encoding

Original Transformer uses fixed sinusoidal patterns:

$$PE_{(pos,2i)} = \sin\left(\frac{pos}{10000^{2i/d}}\right)$$

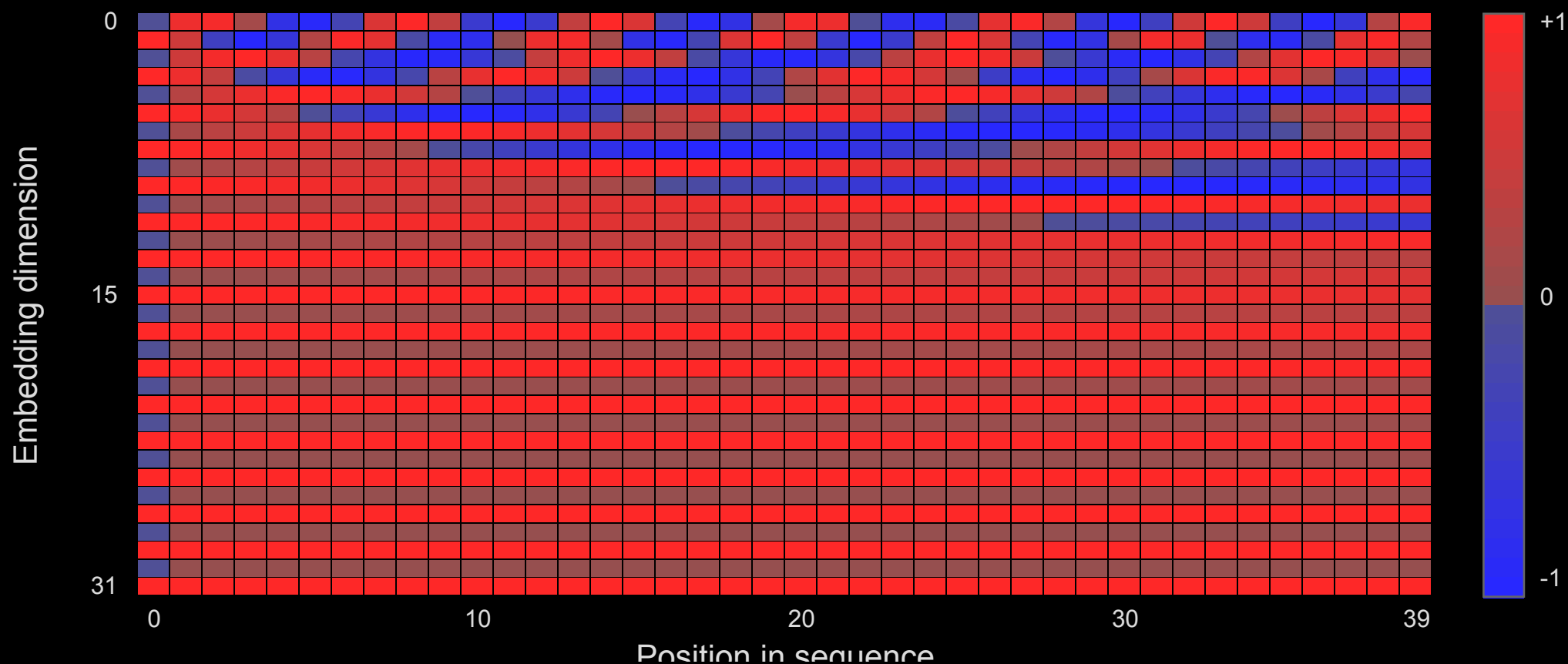
$$PE_{(pos,2i+1)} = \cos\left(\frac{pos}{10000^{2i/d}}\right)$$

Where:

- pos = position in sequence (0, 1, 2, ...)
- i = dimension index
- d = model dimension

Visualizing Positional Encoding

Sinusoidal Positional Encoding: Each Position Has Unique Pattern



Why Sinusoids Work

Properties:

1. **Unique for each position:** No two positions have identical encoding
2. **Smooth variation:** Similar positions have similar encodings
3. **Relative positions:** PE_{pos+k} can be written as linear function of PE_{pos}
4. **Extrapolation:** Works for positions not seen during training

Adding Position to Input

Simply add positional encoding to token embedding:

$$\text{input} = \text{token_embedding} + \text{positional_encoding}$$

Position + Content = Complete Input



Learned vs. Fixed Encodings

Fixed (sinusoidal):

- Original Transformer
- No extra parameters
- Generalizes to longer sequences

Learned:

- BERT, GPT use learned position embeddings
- One learnable vector per position
- May capture more complex patterns
- Limited to maximum sequence length

Both work well in practice!

What We've Covered

In this video, we've learned:

- Self-attention is position-blind without positional encoding
- Sinusoidal encoding: unique pattern for each position
- Different frequencies capture different scales of position
- Add positional encoding to token embeddings
- Learned encodings work too (BERT, GPT)
- Position information enables order-aware processing